

Number system Operations on Rational and Irrational

Hot Shots Playlist

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I = irrational, R = rational

Addition

$I + I$ → I: $\sqrt{2} + \sqrt{3} = \sqrt{2} + \sqrt{3}$

$3 + \sqrt{2} + 3 - \sqrt{2} = 6$

$I + I$ → R: $\sqrt{2} + (-\sqrt{2}) = 0$

$I + R$: $\sqrt{2} + 5 = I$, $\sqrt{2} + 0 = I$

Only one possibility: irrational (Always irrational)

Multiplication

$I \times I$ → I: $\sqrt{2} \times \sqrt{3} = \sqrt{6}$

$I \times I$ → R: $\sqrt{2} \times \sqrt{2} = 2$

$I \times R$ → I: $\sqrt{2} \times 2 = 2\sqrt{2}$

$I \times R$ → R: $\sqrt{2} \times 0 = 0$

Subtraction

$I - I$ → I: $\sqrt{3} - \sqrt{2} = \sqrt{3} - \sqrt{2}$

$I - I$ → R: $\sqrt{2} - \sqrt{2} = 0$

$I - R$: $\sqrt{2} - 5 = I$, $\sqrt{2} - 0 = I$

Only one possibility: irrational (Always irrational)

Division

$I \div I$ → I: $\sqrt{2} \div \sqrt{3} = \frac{\sqrt{6}}{3}$

$I \div I$ → R: $\sqrt{2} \div \sqrt{2} = 1$

$I \div R$: $\sqrt{2} \div 2 = \frac{\sqrt{2}}{2} = I$, $\sqrt{2} \div 0 =$ undefined

If $R \neq 0$, only one possibility: irrational

$R \div I$: $2 \div \sqrt{2} = \sqrt{2} = I$, $0 \div \sqrt{2} = 0 = R$

If $R \neq 0$: irrational; if $R = 0$: rational

R & R ⟶ Always Rational (except $\frac{0}{0}$)

$\sqrt[3]{7}$, $\sqrt{5}$ ← I & I ⟶ Always both possibilities (I or R)

$\frac{R}{I} = \frac{0}{\sqrt{2}} = 0$

✓ ✗

I & R ⟶ Always Irrational (Except in multiplication, and division due to 0)

$\sqrt{3}+5$ $\frac{\sqrt{3}}{5}$ $\sqrt{3}-1$

R = Rational Number I = Irrational Number

$\sqrt{2} \times 0 = 0$

I × R = R

$\sqrt{2} \times 5 = 5\sqrt{2}$

I R = I

1 What type of number is the result of multiplying a Rational Number with an Irrational Number?

a) Always Rational

b) Always Irrational

c) Always Integer

d) Sometimes Rational, Sometimes Irrational

एक परिमेय संख्या को एक अपरिमेय संख्या से गुणा करने पर किस प्रकार की संख्या प्राप्त होती है?

a) सदैव परिमेय

b) सदैव अपरिमेय

c) सदैव पूर्णांक

d) कभी परिमेय, कभी अपरिमेय

2. The product of two rational numbers is always:

a) Rational

b) Irrational

c) Natural

d) Integer

R R

दो परिमेय संख्याओं का गुणनफल हमेशा होता है:

a) परिमेय

b) अपरिमेय

c) प्राकृतिक

d) पूर्णांक

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3. Rational number + Irrational number = ?

A. Rational number

B. Irrational number

C. Either Rational number or Irrational number

D. Integer

परिमेय संख्या + अपरिमेय संख्या = ?

a) परिमेय संख्या

b) अपरिमेय संख्या

c) दोनों में से एक परिमेय संख्या या अपरिमेय संख्या

d) पूर्णांक

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4 What type of number is the result of sum of two Irrational Numbers?

a) Always Rational

b) Always Irrational

c) Always Integer

d) Sometimes Rational, Sometimes Irrational

दो अपरिमेय संख्याओं के जोड़ से किस प्रकार की संख्या प्राप्त होती है?

a) सदैव परिमेय

b) सदैव अपरिमेय

c) सदैव पूर्णांक

d) कभी परिमेय, कभी अपरिमेय

5 If x is irrational and y is irrational, then $x + y$ is:

अगर x अपरिमेय है और y अपरिमेय है, तो $x + y$ है:

a) Always irrational

b) Always rational

c) Rational if $x = -y$

d) Never rational

$\sqrt{2} + (-\sqrt{2}) = 0$

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